UTM
UNIVERSITI TEKNOLOGI MALAYSIA

COURSE INFORMATION

School/Faculty:	**Faculty Computing**		
Program name:	**Bachelor in Software Engineering with Honours**		
Course code:	**SCSM2223**	**Academic Session/Semester:**	**2025/2026/2**
Course name:	**Cross-Platform Application Development**	**Pre/co requisite (course name and code, if applicable):**	**SCST1143 Database**
Credit hours:	**3**		

Course synopsis	This course provides students with a foundation on the development of cross-platform applications. It covers the workflows, tools and frameworks required to develop applications for various computing devices including mobile and desktop platforms. The course adopts current Web technologies as a basis for teaching the process of cross-platform application development. Students are exposed to the Web system architecture, front-end development for user interfaces, backends integration to support service/model, and multi-platform desktop/mobile web development. This course incorporates a project-based learning approach where students are required to complete a given case-study to practice their skill on cross-platform development. At the end of course, the students will be equipped with the competency of the appropriate skills for the development of cross-platform applications as well as personal and entrepreneurial skills.			
Course coordinator (if applicable)	Assoc. Prof. Dr Hishammuddin bin Asmuni@Hasmuni			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Assoc. Prof. Dr Hishammuddin bin Asmuni@	N28a,		hishamudin@utm.my
	Dr. Mohd Razak bin Samingan	N28, FC		mrazak@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO*	PLO (MQF Cluster Code)	Taxonomies and Graduate attributes	T&L methods	Assessment methods
CLO1	Explain and implement the concepts and components of cross-platform web application development, including client-server architecture, frontend frameworks, backend APIs, database integration, and deployment processes.	PLO1 (C1)	C3 (SC2)	Active learning, Problem Solving	LE (20%), ASG (15%), FE (30%)
CLO2	Develop an innovative solution for a given problem by using appropriate cross-platform technologies.	PLO3 (C3A)	C6 (SC3)	Design thinking, Project - problem based learning	PR2 Interim (10%) PR3 Final (15%)

CLO3	Use entrepreneurial mindset in creating solutions for the identified problems , demonstrating resourcefulness in selecting cross-platform technologies, integrating third-party services, and deploying to cloud platforms with mobile adaptability.	PL10 (C4B)	A3 A(ES3)	Project - problem based learning	PR1 Project Proposal (5%) PR4 Presentation (5%)
PR – Project; LE - Lab Exercise; ASG - Assignment ; FE -Final Exam.					

<MQF Cluster Code>
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numercy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism
<Taxonomies>
Cognitive (C) - C1-C6, **Psychomotor (P)** - P1-P7, **Affective (A)** - A1-A5
<Graduate Attributes>
SC1, SC2, SC3 = Scholars, **SI1, SI2, SI3, SI4** = Social Intelligence, **IT1, IT2, IT3** = Innovative Talent, **CC1, CC2, CC3, CC4** = Constructive Citizen, **ES1, ES2, ES3, ES4, ES5** = Enterprising Skills,

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active Learning	Conducted activity to engage students in a new topic, activate prior knowledge, share unit objectives, and monitor students' learning.
2	Design thinking	Conducted collaborative group activity where students effectively teach each other (with the teacher's guidance) a given skill or procedure, topic or problem.
3	Project - case study	Conducted based on given real world problem. The students must manage their projects by submitting specified deliverables based on the given problem. The problems are given in sequential steps throughout the semester.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :
Team Working and Entrepreneurial skills during project execution.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			

Week 1	**Chapter 1: The Web Landscape and Developer Toolkit** **1.1 Introduction to Cross-platform Development** **1.2 Web Applicaton Architecture** Client-server Model, Request-response cycle, Roles of Frontend, Bankend, Database, and API	CLO1					3					2	5
Week 2	**Chapter 2: Buildng the User Inerface** HTML Fundamentals, CSS Styling, Layout Techniques, Responsive Web Design * Lab Exercise 1 (4%)	CLO1					3				2		5
Week 3	**Chapter 3: Making it Dynamic (Javascript & DOM)** JavaScript, DOM Seletion and Manipulation, Event Handling * Lab Exercise 2 (4%)	CLO1, CLO3	3								2		5
Week 4	**Chapter 4: Fetching Data (AJAX, JSON, Fetch API)** Undestanding AJAX, JSON, Fetch API, JQuery * Lab Exercise 3 (4%) * Project Discovery (Initiate)	CLO1, CLO3	3								2	2	7

Week 5	**Chapter 5: Mastering Asynchronous Javascript** Javascript Event Loop, Callbacks, Promises, Asyn/Await * Lab Exercise 4 (4%)	CLO1, CLO2	3								2	2	7
Week 6	**Chapter 6: Frontenf Framework - Vue.js** Why Frameworks? Vue.js overview Vue Instance and Templating Core Directives	CLO1, CLO2	3								2	1	6
Week 7	**Chapter 7: Building Full Vue Apps (CLI, Router, State)** Vue CLI Single File Components (SFCs) Component Communication Vue Router State Management * Lab Exercise 5 (4%)	CLO1, CLO2	3								1	2	6
Week 8	**MID SEM BREAK**												
Week 9	**Chapter 8: Back-end Development** Connecting Vue to Backend Axios HTTp Client Form Handling in Vue JSON Server - performing CRUD operations	CLO1, CLO2	3								2	1	6

Week 10	**Chapter 9: Server-Side and PHP Slim** Server-Side Role RESTful API Design PHP Slim framework Project Interim Progress Routing in Slim Response Formatting * Assignmnet 1	CLO1, CLO2	3								2	1	6
Week 11	**Chapter 10: Connecting to Database** PHP Data Objects (PDO) Implementing CRUD operations Prepared Statement	CLO1, CLO2					3				1	1	5
Week 12	**Chapter 11: Securing the API with JTW** Authentication Concepts JSON Web Tokens (JWT) Implementing JWT authentication	CLO1, CLO2	3								2	1	6
Week 13	**Chapter 12: Backend Security and Best Practices** SQL Injection Cross-site scripting (XSS) Data Tampering Authentication and Authorization * Assignment 2	CLO1, CLO2	3								1	2	6
Week 14	**Chapter 13: Integration, Deployment and Mobile** Full-stack integration & Cloud Deployment Web to Mobile Cross Platform Development	CLO1, CLO2	3								1	2	6
Week 15	**Project Demo** **Project Finale**	CLO2, CLO3				3							3
											SUB-TOTAL SLT :		**79**

Continous Assessment		%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Lab Exercise 1	4			2		2
2	Lab Exercise 2	4			2		2
3	Lab Exercise 3	4			2		2
4	Lab Exercise 4	4			2		2
5	Lab Exercise 5	4			2		2
6	Assignment 1	8			4		4
7	Assignment 2	7			4		4
8	Project Discovery	5	0.5		3		3.5
9	Project Interim	10	0.5		3		3.5
10	Project Presentation/Demo	20	1		3		4
					SUB-TOTAL SLT :		**29**

Summative Assessment		%	Face-to-Face (F2F)		NF2F Independent Learning for		TOTAL SLT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Final Exam	30	3			9	12
					SUB-TOTAL SLT :		**12**
					SLT for Assessment:		**41**
					GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	31.67
B	% SLT for Online & Independent Learning Component :	68.33
C	%SLT for Online Component:	46.67
D	% SLT for All Practical Component:	0.00
D1	% SLT for F2F Physical Practical Component:	0.00
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab,
Software: Laragon, Sublime Text, Sublime Merge, Github, Capasitor

References (include required and further readings, and should be the most current)
Books Freeman, E., & Robson, E. (2020). Head First HTML and CSS (2nd Edition). O'Reilly Media. ISBN: 978-1492073406. Scott, A. D. (2020), “JavaScript Everywhere : Building Cross-platform Applications with GraphQL, React, React Native, and Electron”, O'Reilly Media. Au-Yeung, J. (2021), “Vue.js 3 By Example : Blueprints to learn Vue web development, full-stack development, and cross-platform development quickly”, Packt Publishing Limited. A**dditional references** W3C. (n.d.). HTML & CSS Standards. World Wide Web Consortium (W3C). https://www.w3.org/standards/webdesign/htmlcss JavaScript.info. (n.d.). The Modern JavaScript Tutorial. https://javascript.info/ REST API Tutorial. (n.d.). What is REST? https://restfulapi.net/ Vue.js. (n.d.). Vue.js Documentation. https://vuejs.org/guide/introduction.html Slim Framework. (n.d.). Slim Framework Documentation. https://www.slimframework.com/docs/ **Learning Management System (LMS)** http://elearning.utm.my

Other additional information (if applicable)
Academic honesty and plagiarism: *(Below is just a sample)*
Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else’s work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

1. To pass this course, students must pass both Continuous and Final assessment.

2. All students must adhere to the following to pass a computing course:
i. Pass both continuous assessment AND final assessment
ii. Get at least minimum pass marks following UTM regulation (40%)
- The passing mark for Continuous Assessment is 30% of continuous marks.
- The passing mark for the Final Assessment is 20% of the final marks.-
- If a student does not achieve the passing mark for continuous assessment or the passing mark for the final assessment, then the student's overall final mark will be set below the UTM passing mark.
Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
3. Students are required to behave and follow the University's dressing regulation and etiquette all the time.
4. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
5. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.
6. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

Disclaimer:

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While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

Prepared by:	**Certified by:**
Name: Prof. Madya Dr Hishammuddin bin Asmuni	**Name: Prof. Madya Dr Mohd Adham bin Isa**
Signature: 	**Signature:** 
Date: 15-Mar-26	**Date:** 16 March 2026